

Selenium

Continuous testing helps to verify the functional aspects of an application in an automated manner and keeps it production-ready after all verifications. Selenium is the most popular automated functional testing tool that many organisations utilise effectively for Web automation.

Selenium characteristics

Open source	Yes
Licence	Apache License 2.0
Type	Automation testing
Written in language(s)	Selenium Server: Java; Selenium WebDriver (works without Selenium Server), official support: JavaScript (Node.js), Python, Ruby, Java or C#
Operating system supported	Cross-platform
Stable release	3.141.59
Website	https://www.selenium.dev/
Companies and projects that use Selenium	Trivago, Cred, HubSpot, OLX Group

A few important Selenium commands

Description	Command
Browser initialisation	<pre>WebDriver driver = new FirefoxDriver(); WebDriver driver = new ChromeDriver(); WebDriver driver = new InternetExplorerDriver();</pre>
Desired capabilities	<pre>DesiredCapabilities caps = new DesiredCapabilities(); caps.setCapability("browserName", "browser"); caps.setCapability("browserVersion", "version"); caps.setCapability("platformName", "platform"); WebDriver driver = new ChromeDriver(caps);</pre>

Selenium is extremely popular for application functional testing in an automated manner.

ELK Stack

ELK is the popular acronym for three open source projects:

- *Elasticsearch*: Search and analytics engine built with RESTful APIs
- *Logstash*: Server-side data processing pipeline that collects data inputs and provides these to Elasticsearch
- *Kibana*: Visualisation of data with charts and graphs

ELK Stack characteristics

Open source	Yes
Licence	Apache License 2.0
Type	Logging, search and index
Written in language(s)	Java
Operating system supported	Cross-platform
Website	https://www.elastic.co/
Companies and projects that use ELK Stack	Netflix, Stack Overflow, LinkedIn

ELK Stack is more popular in Kubernetes environments, where you can utilise it for effective logging mechanisms in Kubernetes clusters installed and configured in cloud environments such as AWS and Azure.

Nagios

Nagios is an open source monitoring tool with a client-server architecture. It is relatively scalable, manageable and secure, and comes with an informative dashboard, log and database system.

Nagios characteristics

Open source	Yes
Licence	GPLv2
Type	Monitoring
Written in language(s)	C
Operating system supported	Cross-platform
Stable release	4.4.6
Website	https://www.nagios.org/
Companies and projects that use Nagios	Airbnb, Cisco, PayPal

It helps to monitor services such as HTTP, SMTP, HTTP, SNMP, FTP, SSH, POP, and so on.